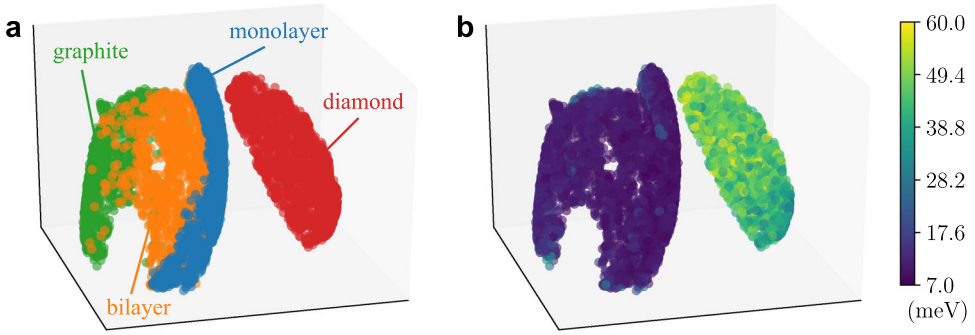


Fig. 4 Representations of local atomic environments in carbon systems. The representations are obtained by embedding the local atomic environments of atoms using the uniform manifold approximation and projection (UMAP) method. Each dot in the plot is the UMAP-embedded descriptor representation of the environment of a single atom. The dots are colored according to **a** their parent structure (as indicated in the figure), and **b** the uncertainty in atomic energy (with the color coding indicated by the legend). Note that although monolayer and bilayer graphene and graphite configurations were included in the training set of the DUNN potential, the specific environments plotted in the figure are drawn from configurations that were not in the training set.

MD trajectories. As noted above, the DUNN potential is trained against monolayer and bilayer graphene and graphite, but the atomic environments we consider here are also drawn from a set of diamond configurations not included in the training set. Our intention is to test whether the uncertainty increases for environments that are "far" from the training set. Specifically, since no diamond-related configurations are included in the training set, we expect the uncertainty for environments drawn from perturbed diamond structures to be larger than other environments that are closer to the training set. To visualize this, we apply the uniform manifold approximation and projection[46] dimensionality reduction technique to embed the atomic environment descriptors into a three-dimensional space. The sampled environments, color coded by their parent structure, are shown in Fig. 4a. We see that the four carbon allotropes form continuous clusters that are separated from each other.

The corresponding uncertainty in the atomic energy (energy of an individual atom) is plotted in Fig. 4b. As anticipated, the uncertainty in the energy for atoms in diamond configurations is much higher than for those drawn from monolayer, bilayer, and graphite configurations. We note that all four carbon allotropes have a similar cohesive energy of ~8 eV/atom; therefore, it is reasonable to compare absolute energy uncertainties instead of relative uncertainties. For a more quantitative comparison, Fig. 5a presents a histogram of the uncertainty in atomic energy for all sampled environments. The uncertainty for environments near the training set (monolayer and bilayer graphene and graphite) is centered at ~10 meV, whereas for diamond environments it is more than four times larger at 43.9 meV. To verify that this difference is indeed due to distance from the training set, we refit the DUNN potential, this time including perturbed diamond configurations in the training set. The histogram obtained using the new potential is plotted in Fig. 5b. The uncertainty in the monolayer, bilayer, and graphite configurations hardly changes, whereas the uncertainty for the diamond environments decrease significantly to the same level as the other carbon allotropes. This provides strong confirmation that on average the uncertainty estimate is able to detect whether configurations are in or out of the training set. Also worth mentioning is the shape of the histogram. The diamond histogram for the original potential (Fig. 5a) is very close to a normal distribution, whereas for the new potential with diamond in its training set, the histogram becomes skewed with a heavier tail on the larger uncertainty side (similar to the other allotropes) (Fig. 5b).

These results support the notion that the DUNN estimate for the uncertainty in atomic energy correlates with distance from the training set. This means that by comparing the energy uncertainties of atomic environments associated with a QOI with the average uncertainty associated with the training set environments, it is possible to determine whether or not a DUNN potential is suitable for that QOI. This can also be used as a criterion for when certain configurations needs to be added to the training set in order to obtain a reliable estimate for a desired QOI.

To place the above results in the context of existing methods, it is of interest to compare them with an ensemble model approach. As discussed in the introduction, ensemble models (e.g., committee models constructed using training set subsampling) have been widely used to quantify uncertainty in machine learning IPs. We revisit the above transferability limits analysis for a committee model to contrast its performance with that of the DUNN potential. (See Supplementary Note 5 for technical details on the committee model). The histogram of the uncertainty obtained using the committee model (Supplementary Fig. 3) is qualitatively similar to Fig. 5. Both approaches are able to determine transferability limits since the histogram of the uncertainty for the training set (monolayer, bilayer, and graphite) does not overlap with the histogram for configurations associated with the QOI (diamond). However there are important distinctions. The distributions obtained from the committee model are far broader, and perhaps due to the limited sample size, the committee model predicts an uncertainty for diamond configurations that is about an order of magnitude larger than the DUNN potential when the training set does not contain diamond configurations (cf. Fig. 5a with Supplementary Fig. 3a). A quantitative comparison of the uncertainty obtained using the two models is provided in the next section ("Precision versus accuracy"). A final difference is the computational cost. In order to obtain good statistics, a minimum of 20 NNs had to be included in the committee model. This means that training the committee model is 20 times more expensive than training a DUNN potential in this case. This is acceptable for one-pass type training, but it can become prohibitive for more sophisticated iterative learning approaches such as active learning.

Precision versus accuracy

Estimates for prediction uncertainty are important for determining when an IP can be trusted. However, low uncertainty does not necessarily mean that a prediction is close to reality. It may seem that it is not possible to know the error in a prediction without access to more accurate calculations or experimental results; however, under certain conditions, uncertainty can provide an estimate. These questions are tied to notions of precision and accuracy.

Given a set of predictions obtained by varying an IP's parameters, accuracy refers to the difference between the mean

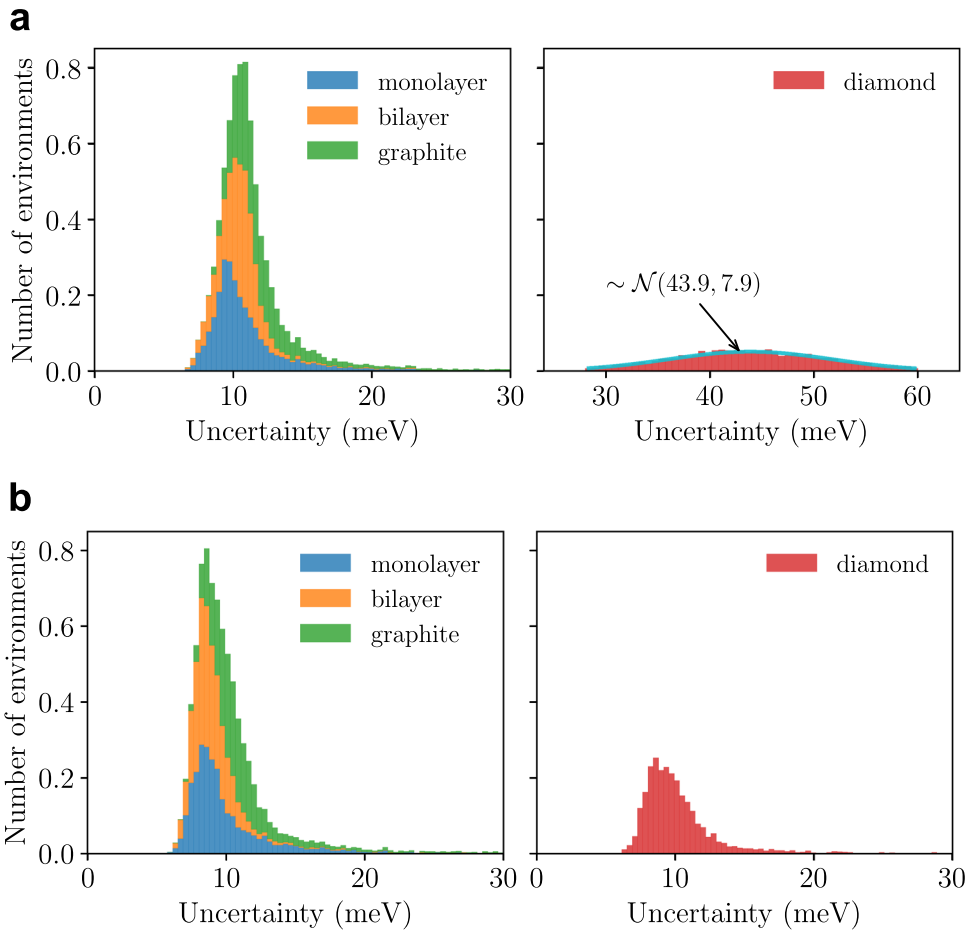


Fig. 5 Histogram of the uncertainty in atomic energy. In **a**, the training set consists of monolayer and bilayer graphene and graphite, but not diamond, while in **b**, the training set includes all four allotropes. The cyan curve in the right plot of **a** represents a normal distribution fitted to the histogram. For each carbon allotrope, 4000 local atomic environments are randomly selected. The vertical axis is normalized so that the area for each carbon allotrope integrates to one, and the histograms on the left are stacked.

prediction and the exact value (e.g., DFT result), and precision refers to the spread in the predictions (i.e., the uncertainty). Recent work by Sethna and co-workers[47,48] has shown empirically that in certain cases accuracy and precision are correlated. This is important because it means that a measure of uncertainty (such as that provided by the DUNN potential) can also be used to estimate the accuracy of a prediction even when the exact values are unavailable.

To study the relationship between accuracy and precision in the predictions of a DUNN potential, we begin by considering the energy of monolayer graphene as a function of the in-plane lattice parameter a. At each value of a, the sampling method is used to obtain the predictive mean and uncertainty from a set of energy calculations. The DUNN results along with DFT data are plotted in Fig. 6a. The energy of the graphene increases with distance from the equilibrium value of $a = 2.466$ Å. A more explicit comparison of accuracy and precision is given in Fig. 6b, where we plot the prediction error (difference between the DUNN mean value and DFT result) and the uncertainty band. For lattice parameters in the range $a = (2.37, 2.57)$ Å that fall within or very close to the training set, the prediction error is bounded by the uncertainty, which is ~10 meV/atom in agreement with the box plot in Fig. 2 and on the same level observed in Fig. 5. As configurations get further from the training set, both the prediction error and uncertainty grow rapidly. For configurations that are not "too far" from the training set ($a \in (2.34, 2.37)$ Å and $a \in (2.57, 2.63)$ Å), the uncertainty continues to provide a bound on the error. However, beyond these values the uncertainty underestimates the prediction error.

The above results suggest that energy uncertainty provides a bound on prediction error for configurations whose uncertainty falls within the training set distribution. To test this heuristic, we examine the energy accuracy and uncertainty for all configurations in the training and test sets. (We also studied accuracy versus uncertainty for forces, see Supplementary Note 6 for details.) The energy prediction error is defined as the absolute value of the energy residual (i.e., the difference between the energy predicted by the DUNN potential and the DFT reference energy). The uncertainty is computed using the sampling method. The prediction error versus uncertainty relation is presented in Fig. 7. Focusing on Fig. 7a that shows all configurations, we see that there is a general trend for the error to increase with uncertainty, and although there is a great deal of scatter, the uncertainty bounds the error for most configurations (points in the shaded region have smaller error than uncertainty). To understand the exceptions, consider that our dataset contains two types of configurations: (1) perfect crystals and (2) perturbed configurations drawn from MD trajectories where the atoms are off-lattice. These two configuration types are plotted separately in Fig. 7b, c. For perfect crystal configurations (Fig. 7b), the error is bounded by the uncertainty for all configurations in agreement with the biaxial straining results presented in Fig. 6. Thus, cases where the uncertainty fails to bound the error are associated with perturbed configuration as shown in Fig. 7c.